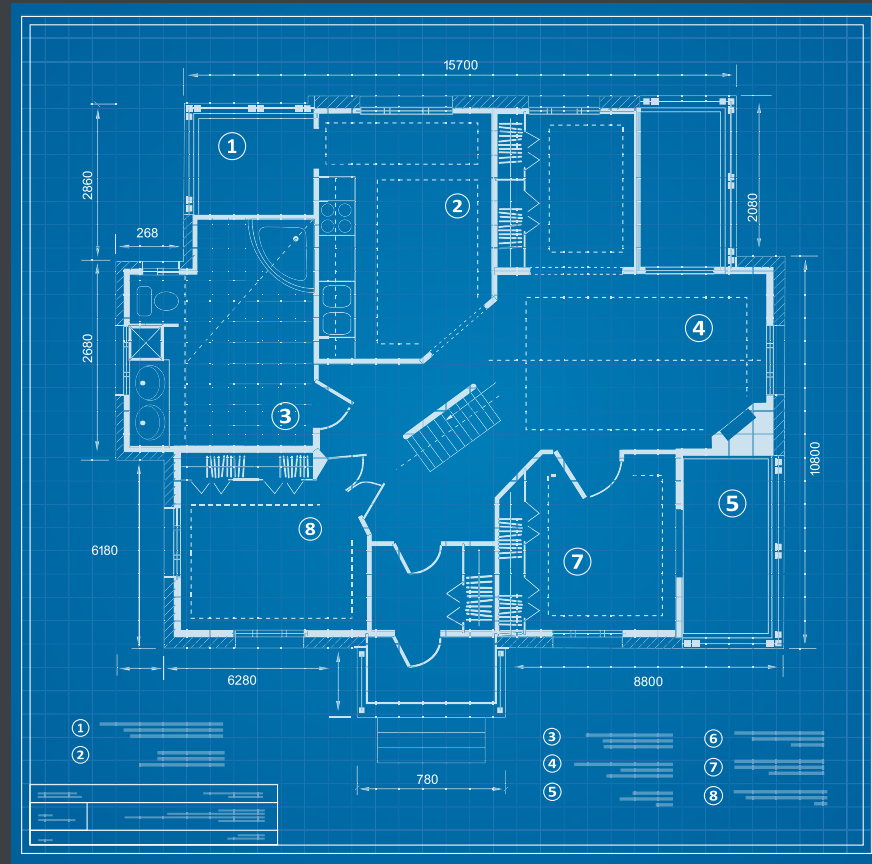


WHAT IS ESTIMATING?

WHAT IS ESTIMATING?

- Process of forecasting costs required to complete a construction project
- Quantifying resources (materials, labour, equipment and other resources) and assigning costs

ESTIMATING



Goal: Accurately calculate the costs to complete the project



WHY DO WE NEED TO ESTIMATE?

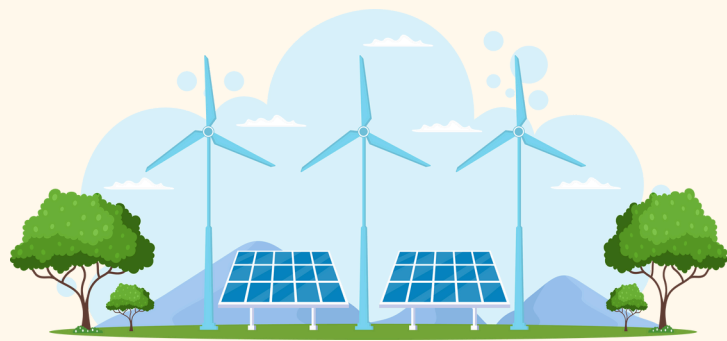
Feasibility

determine how much the project will cost before we commit to it.

Bidding/Quoting
prepare competitive submissions

Cost Control
inform budgets and forecasts

WHO NEEDS ESTIMATES?



Project Developers



**Head-
Contractors**



Sub-contractors



Banks/Institutions

TYPES OF ESTIMATES

- **Order of magnitude:** $\pm 30-50\%$, based on minimal design
- **Budget estimate:** Based on schematic design, $\pm 20\%$
- **Detailed estimate:** Based on full construction documents, $\pm 10\%$
- **Definitive/Tender estimate:** Final pricing used for contract

WHY ESTIMATING MATTERS

THE REALITY OF THE CONSTRUCTION INDUSTRY

- Commoditized industry
 - Clients choose contractors based on price
 - Drives down profit margins
- Huge amount of risk
 - 30% of all business insolvencies are in construction
- Highly contractual
 - Contracts with huge risk transfer and strict terms and conditions
- Complex and technical scope
 - Every project is different

THE REALITY OF THE CONSTRUCTION INDUSTRY

- What this means
 - The lowest bidder typically wins
 - One bad project can send your business broke
 - Your offer consists of two key parts
 - Your price
 - Your terms and conditions (the contract)

THE ROLE OF THE ESTIMATOR (AND WHY IT MATTERS)

To secure work at a price we can
deliver the project (uphill battle)

**More than plugging numbers into a
spreadsheet**



ESTIMATING IS AN ART - NOT A SCIENCE

- Two estimators can price the same job, using the same drawings and come up with different prices
 - And both be 100% correct
- Estimates depend on assumptions; productivity rates, programs, risk, procurement strategies, subcontractor relationships and contracts
- There is no “correct” price - only a price that makes sense for your business that maximises your chances of **securing profitable work**

ESTIMATING IS A BUSINESS DECISION

- Your estimate is a function of:
 - How much your company wants to win the project
 - How much risk you are willing to carry
 - What profit margin you are targeting
 - How you can spread overheads
 - What terms, conditions and scope you are willing to accept
- Accurately pricing the project is not the goal (it's a step)
- Winning the project at a price you can deliver is the goal

ESTIMATING IS THE BASIS OF CONTRACTS

- When you submit an estimate - you're not just "calculating the costs", you're making a contractual commitment
- This means any estimate has two core components
 - The price
 - The terms and conditions
- Your terms and conditions are just as important as your price
 - Scope of works
 - Exclusions
 - Qualified assumptions

ESTIMATING MATTERS BECAUSE

- The construction industry makes decisions on price
- Price too high - never win work
- Price too low - win work and lose money in delivery
- Estimating is how you find the right balance between the two to **win profitable work**

UNDERSTANDING CONSTRUCTION COSTS

UNDERSTANDING CONSTRUCTION COSTS

- Foundation to understand estimating
- Where we spend money to projects

ESTIMATE - COMPONENTS

Sell Price - Margin,
Corporate
Overheads, R&O

Total Project Costs

Direct Costs

In-Direct

Labour

Materials

Re-curring

Non-Recurring

Plant

Sub-Contract

DIRECT COSTS

- Directly tied to completing physical work on site
- Measurable and typically vary based on the scope of job
 - I.e. bigger concrete foundations need more concrete
- Four main types
 - Labour
 - Materials
 - Plant & Equipment
 - Subcontractor costs

DIRECT COSTS - LABOUR

- Includes wages, allowances, on-costs (super, workers comp)
- Labour productivity (e.g. m^2/day or m^3/hour) drives total cost
- Can vary significantly based on
 - skill level
 - Work type
 - Project location

DIRECT COSTS - MATERIALS

- Raw inputs (e.g. concrete, rebar, electrical cables)
- Includes supply, freight and wastage
- Volatile and often subject to escalation

DIRECT COSTS - PLANT & EQUIPMENT

- Hired or owned equipment needed for construction (cranes, excavators, concrete pumps)
- Includes mobilisation, demobilisation, fuel, servicing and operator costs
- Needs to align with productivity assumptions

DIRECT COSTS - SUBCONTRACTOR COSTS

- Quoted prices for packages of work (e.g. electrical, plumbing, formwork)
- Often appear as lump sums but still need to be validated against quantities and scope
- May include risk-sharing or exclusions that must be accounted for

ESTIMATE - COMPONENTS

Sell Price - Margin,
Corporate
Overheads, R&O

Total Project Costs

Direct Costs

In-Direct

Labour

Materials

Re-curring

Non-Recurring

Plant

Sub-Contract

INDIRECT COSTS

- Costs required to run the project but not tied to individual work items
 - Site supervision
 - Site office set up and operation
 - Temporary works - scaffold, hoardings, access stairs
 - Site utilities
 - Safety compliance
 - Permits and approvals
- Typically form a large portion of overall project costs (~30%)

INDIRECT COSTS

- Re-curring
 - On-going costs throughout the project lifecycle
 - I.e. staff salaries
 - Driven by the overall duration of the project
- One-off costs
 - Incurred at a single point in time
 - I.e. site mobilisation and freight costs

TOTAL PROJECT COSTS

Direct Costs + In-Direct Costs = Total Project Cost

What would it cost you to complete the project based on your understanding of the scope and assumptions

CONTINGENCY & RISK

- Used to cover unknowns — things that might happen but aren't guaranteed
- Should be based on risk registers or historical data
- Can be broken out per item or included as a % at the end
- Depends on clarity of scope, contract terms, design information, ground conditions, etc.

MARGIN

- Business decision - not just a cost
- Should reflect
 - Overall risk (unknown unknowns)
 - Competition
 - Desired profit
 - Corporate overheads
- In competitive markets this will be squeezed, need to understand your true costs!

ESTIMATING METHODS

ESTIMATING METHODS

- There are two main ways to calculate costs
 - Top-down (budget) estimating
 - Bottom-up (first principles) estimating
- Vary based on
 - Accuracy
 - Level of effort

TOP-DOWN

- A quick, high-level method of estimating based on costs from similar past projects
- Rather than breaking down every individual cost, you apply unit rates or multipliers to the overall project scope

If Project A costs \$10 million and Project B is 50% smaller, we assume Project B will cost around \$5 million - subject to adjustments

TOP-DOWN

When to Use It:

- Feasibility studies
- Early-stage client discussions
- Rough-order-of-magnitude pricing
- When the design is not finalised
- When you're scoping whether to pursue a project or not

TOP-DOWN

When to Use It:

- Feasibility studies
- Early-stage client discussions
- Tendering projects that are not yet live
- When you're scoping whether to pursue a project or not
- Rough-order-of-magnitude pricing before detailed pricing (checking detailed pricing)

TOP-DOWN

Advantages

- Fast and easy to prepare
- High-level decision making
- Verifying bottom-up estimates are correct

TOP-DOWN

Limitations

- Lacks detail and misses unique scope differences
- Relies heavily on historical accuracy and similarity
- Not suitable for formal bids or contracting purposes

BOTTOM-UP

- Building the project costs from the ground up
- Breaking project down into activities and estimating the costs individually
- Process
 - Create WBS
 - Price individual activities
 - Sum the direct costs together
 - Estimate indirect costs
 - Apply profit and overheads

BOTTOM-UP

- When to use it
 - Tender submissions
 - Detailed cost planning
 - Contract award pricing
 - Anytime you need accuracy, traceability and defensibility

BOTTOM-UP

- Advantages
 - High accuracy and control
 - Transparent and traceable
 - Allows better analysis and change management
 - Supports confidence

BOTTOM-UP

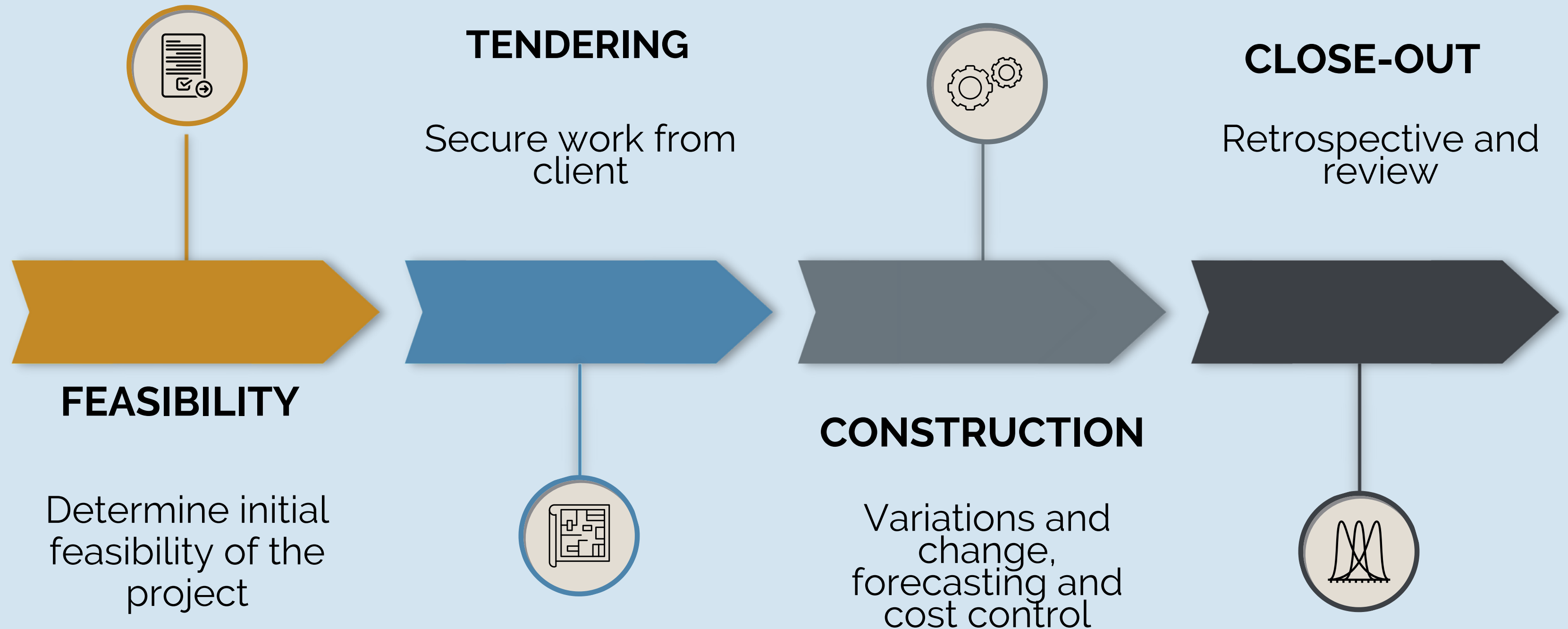
- Limitations
 - Time-consuming and resource-intensive
 - Requires detailed design and scope clarity
 - Demands more skills and expertise

COMBINE THE TWO

- Initial RoM pricing based on budget approach
- Detailed bottom-up pricing for contract award

WHEN TO ESTIMATE

ESTIMATING DURING THE PROJECT LIFECYCLE



FEASIBILITY STAGE: MAKING THE BUSINESS CASE

Purpose: Determine if a project is feasible and worth pursuing. By developers.



FEASIBILITY STAGE: MAKING THE BUSINESS CASE

- Use top-down estimating to produce a rough-order-of-magnitude (ROM) cost.
- Compare to benchmark projects or \$/m² rates.
- Factor in high-level risks, site conditions, and escalation.
- Provide input to:
 - Business case approvals
 - Investor or board presentations
 - Preliminary funding estimates

FEASIBILITY STAGE: MAKING THE BUSINESS CASE

Key Considerations:

- Design is usually incomplete or conceptual
- Accuracy is low
- Used for key decisions and value engineering

TENDERING & PROCUREMENT

Purpose: Submit a competitive, accurate offer that wins the project at a price we can deliver it

TENDERING & PROCUREMENT

- Shift to bottom-up estimating based on a detailed scope of work.
- Conduct quantity takeoffs, calculate direct and indirect costs.
- Obtain subcontractor and supplier quotes.
- Review contract terms and identify risks.
- Apply contingency and set profit margins.
- Produce a submission with:
 - Clarifications
 - Assumptions
 - Value engineering options (if allowed)

TENDERING & PROCUREMENT

Key Considerations:

- This is where contractual commitments are made — scope, price, exclusions.
- Poor estimates here lead to losses, disputes, and delivery issues.
- Strategic pricing decisions (e.g., to secure backlog or market share) must still be grounded in real costs.

CONSTRUCTION

Purpose: Manage costs, track performance, and keep delivery aligned with budget.

CONSTRUCTION

How Estimating Is Used:

- The tender estimate becomes the baseline for budget and cost control.
- Used to set:
 - Package budgets
 - Procurement targets
 - Cash flow forecasts
- Reforecasting as conditions change (e.g., delays, design changes).
- Supports:
 - Variation pricing
 - Subcontractor negotiations
 - Progress claims
 - Forecasting

CONSTRUCTION

Key Considerations:

- A good estimate supports confident decision-making during delivery.
- Poor visibility of original estimating logic leads to budget blowouts.

CLOSE-OUT

Purpose: Compare actual performance to estimated and refine future estimating accuracy

CLOSE-OUT

How Estimating Is Used:

- Conduct cost reconciliation: estimated vs actual cost.
- Identify variances and root causes (e.g. productivity issues, scope creep).
- Feed back into:
 - Historical cost database
 - Productivity benchmarks
 - Risk pricing assumptions
- Lessons learned help improve:
 - Future tenders
 - Strategic pricing
 - Delivery planning

CLOSE-OUT

Key Considerations:

- A robust close-out review can massively improve your win rate and margin on future jobs.
- Capture the learnings while they're fresh — not months later.

TOP-DOWN (BUDGET) ESTIMATING

PROCESS

1. Identify a relevant reference project
 - a. Use a completed or accurately estimated project that matches the scope, location and type
2. Calculate the unit rate
 - a. Convert the total cost of the project into a unit rate (e.g. \$ per m³ of concrete, \$ per lineal m)
3. Scale to the new project
 - a. Multiply the unit rate by the known quantity
4. Adjust for scope differences
 - a. Adjust for known changes to scope, location, and market factors
5. Document assumptions clearly

EXAMPLE

Estimating the costs for a solar farm construction

Reference project: 25MW solar farm



STEP 1: Use Historical Project Data

- A 25MW solar farm was previously delivered for \$30 million
- Unit rate
 - $\$30\text{m} \div 25\text{MW} = \$1.20/\text{W}$

STEP 2: Scale the Cost to the New Project

- New Project is 20MW
- Apply the same unit rate
- $\$1.20/\text{W} \times 20\text{MW} = \24 million

STEP 3: Adjust for Known Scope Changes

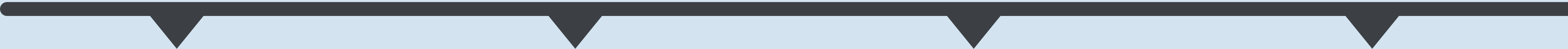
- This project the client will free-issue the inverters
- Cost per inverter on previous project was around \$0.10/W
- Adjust the unit rate
 - $\$1.20/W - \$0.10/W = \$1.10/W$
- Recalculate based on 20MW:
 - $\$1.10/W \times 20MW = \$22m$

STEP 4: Document Key Assumptions

- Based on similar project scope, scale, and site conditions.
- Excludes inverters (client to supply).
- No allowance made for escalation, design variation, or site-specific challenges (to be refined in detailed estimate).

FIRST-PRINCIPLES ESTIMATING

ESTIMATING - PROCESS



DIRECT COSTS

Determine
the cost of
the
deliverables

IN-DIRECT

Determine
requirements
for overheads
and
management

SUMMARISE

Apply risk,
opportunity
and mark-up

SUBMIT

Review,
check for
errors

ESTIMATING - DIRECT COSTS

1. **Work Breakdown Structure** - Decompose project scope into deliverables
2. **Quantity Take-off** - Measure key quantities of work
3. **Estimate** - Determine labour, plant, material and sub-contract costs for each activity
4. **Summarise** - Group costs to calculate total project costs

ESTIMATING - IN-DIRECT COSTS

1. **Identify In-direct Costs** - Identify applicable management and overhead expenses
2. **Recurring In-direct Costs**- Determine project duration (schedule), calculate cost per unit of time
3. **Non-Recurring In-direct Costs** - Calculate costs in same way as direct costs

ESTIMATING - SUMMARISE

1. **Risk and Opportunity** - Identify the uncertain events or conditions impacting the project. Make allowances accordingly.
2. **Margin** - Apply profit margin and corporate overhead allowances

ESTIMATING - SUBMIT

1. **Review estimate** - errors, omissions and accuracy
2. **Letter of offer** - clearly document and detail exclusions, allowances, etc.

DIRECT COSTS

DIRECT COSTS

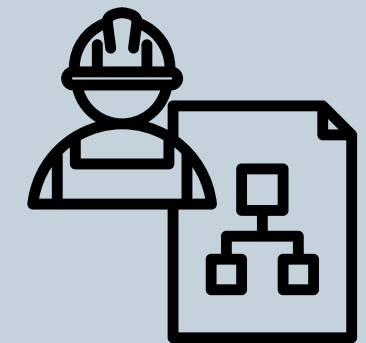
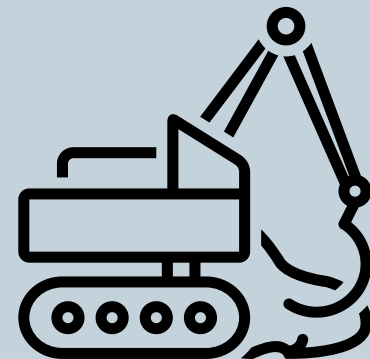
- Direct costs are expenses that can be directly traced to specific construction activities
 - I.e. earthworks, concrete foundations, etc.
- Costs of physically building the project - the costs of “doing the work”
- They vary based on the size, complexity and scope of the project

LABOUR

PLANT & EQUIPMENT

MATERIALS

SUBCONTRACT



DIRECT COSTS

LABOUR

- Wages, allowances and on-costs
- Productivity factors
 - Task productivity - mhr per m2 of formwork
 - Overall productivity factor - i.e. 85%
 - Daily productivity - i.e. 50m of trenching per day for crew of 3
- Crew size and composition - formworkers, steel fixers, electricians

LABOUR

- On-costs contribute significantly to overall labour costs
- On-costs
 - Superannuation
 - Workers compensation insurance
 - Payroll tax
 - Leave entitlements (sick leave, annual leave)
 - Site-specific allowances (e.g. travel, site, tool allowances)

LABOUR

- How to calculate labour costs

1. Base rate

- a. Hourly rate paid to the worker (e.g. \$50/hr)

2. On-costs

- a. Typically 25%-40% of base rate
- b. i.e. $\$50/\text{hr} + 30\% \text{ on-costs} = \$65/\text{hr}$

3. Productivity

- a. Productivity = how much a worker or crew can physically achieve
- b. i.e. A crew of can trench 50m per day
- c. Labour cost per m = $(3 \times 10 \text{ hrs} \times \$65/\text{hr}) / 50 = \$39/\text{m}$

PLANT & EQUIPMENT

- Machinery and equipment used for construction works
 - Cranes, excavators, concrete pumps, scaffolding, etc.
- Mobilisation and demobilisation costs
- Fuel, servicing and maintenance

PLANT & EQUIPMENT

- Hired plant
 - Charged at daily or hourly hire rates
 - Includes operator (wet hire) and basic servicing
 - I.e. Concrete pump = \$1,200/day
- Owned plant
 - Still needs to be charged to project (internal rate)
 - Cost basis includes
 - Depreciation (spread over useful life)
 - Financing costs (interest or lease payments)
 - Servicing and maintenance
 - Fuel and consumables

PLANT & EQUIPMENT

- Key considerations
 - Utilization: idle plant incurs costs
 - Overestimating plant time inflates costs
 - Mob/de-mob costs

SUBCONTRACTOR COSTS

- Lump-sum or unit rate quotes from specialist contractors
 - Electrical, plumbing, structural steel, roofing, etc.
- Subcontractors provide all labour, plant, and materials in their fixed price

SUBCONTRACTOR COSTS

- How to handle subcontractor costs
 - Develop a detailed scope of works
 - Clear inclusions/exclusions
 - Verified quantities
 - Obtain 3-6 quotes
 - Don't rely on the cheapest one
 - Gap analysis
 - Review the quotes and adjust for any exclusions/gaps
 - Perform a high-level first principles check if applicable
 - Include the middle price in your estimate

DIRECT COSTS

Direct Costs = Labour + Plant + Materials + Sub-contract Costs

DIRECT COSTS - EXAMPLE

Scope

- 100m² Concrete slab, 150mm thick



DIRECT COSTS - EXAMPLE

Quantity Take-Off

- Concrete volume: $100\text{m}^2 \times 0.15\text{m} = 15\text{m}^3$ of concrete
- Formwork area (perimeter): $40\text{m} \times 0.15\text{m} = 6\text{m}^2$ of formwork
- Reinforcement: $100\text{kg}/\text{m}^3 \times 15\text{m}^3 = 1,500\text{kg}$ of rebar

Item	Quantity	Rate	Cost
Concrete Supply	15m3	\$250/m2	\$3750
Reinforcement Supply	1,500kg	\$1.50/kg	\$2,250
Formwork Materials	6m2	\$50/m2	\$300
Labour	3 workers x 8 hours	\$60/hr	\$1,440
Concrete Pump	1 day	\$1,200/day	\$1,200
Total			\$8,940

ESTIMATE - COMPONENTS

Sell Price - Margin,
Corporate
Overheads, R&O

Total Project Costs

Direct Costs

In-Direct

Labour

Materials

Re-curring

Non-Recurring

Plant

Sub-Contract

INDIRECT COSTS

INDIRECT COSTS

- Expenses necessary to run the project, but cannot be directly linked to a specific construction activity or work package
- Support the overall project delivery, covering site operations and compliance.
- Also known as preliminaries (prelims) or site overheads

INDIRECT COSTS

- Why indirect costs matter
 - Often underestimated - eating into margins
 - Driven by the project duration
 - Hard to quantify (unlike direct costs) - much more subjective

INDIRECT COSTS - KEY CATEGORIES

- Site Supervision & Management
 - Project Manager, Site Engineer, Foreman salaries or wages.
 - Support staff: document controllers, safety officers.
 - On-costs apply (superannuation, insurance, leave, etc.).
 - Duration-based: linked to program length

INDIRECT COSTS - KEY CATEGORIES

- Site Establishment & Facilities
 - Set up and removal of site compound.
 - Site office purchase/rental, fit-out, utilities (power, water, internet).
 - Amenities: toilets, lunchrooms, first aid stations.
 - Fencing, signage, security systems.

INDIRECT COSTS - KEY CATEGORIES

- Temporary works
 - Access roads, scaffolding, shoring, formwork not part of permanent works.
 - Crane bases, working platforms, hoardings, traffic management setups.
 - These are necessary to carry out works safely and efficiently.

INDIRECT COSTS - KEY CATEGORIES

- Utilities and consumables
 - Power, water, fuel for generators.
 - Consumables: PPE, office supplies, cleaning materials.
 - Waste disposal and environmental compliance measures.

INDIRECT COSTS - KEY CATEGORIES

- Permits, Insurance and Compliance
 - Local council permits, road closures, environmental fees.
 - Project-specific insurance (contract works, public liability).
 - Safety compliance: training, audits, documentation.

INDIRECT COSTS - KEY CATEGORIES

- Mobilisation & demobilisation
 - Transport of equipment and personnel to/from site.
 - Includes mobilisation of plant, cranes, heavy machinery.
 - Setup costs for subcontractors may also be charged to prelims.

INDIRECT COSTS - HOW TO CALCULATE

- Step 1: Estimate Duration
 - Indirect costs are often time-based - longer project = higher prelims
 - Program drives the staffing, site facility needs and utility costs

INDIRECT COSTS - HOW TO CALCULATE

- Step 2: Quantiy Resources
 - Staff: How many people and for how long?
 - Facilities: size, type, rental period
 - Temporary works: what's needed based on site conditions?

INDIRECT COSTS - HOW TO CALCULATE

- Step 3: Apply unit rates
 - Salaries, hire rates, utilities
 - Example
 - Site engineer: \$100k/year for 12 months
 - Office rental: \$2,000/month

INDIRECT COSTS - HOW TO CALCULATE

- Step 4: Add fixed costs
 - Mobilisation, permits, one-off compliance fees
 - Insurance premiums based on contract value

INDIRECT COSTS - KEY CONSIDERATIONS

- Time overruns directly increase indirect costs - plan conservatively
- Indirect costs are non-negotiable: even if direct costs are cut, these remain
- Some indirect costs may be client reimbursable - check contract terms

ESTIMATE - COMPONENTS

Sell Price - Margin,
Corporate
Overheads, R&O

Total Project Costs

Direct Costs

In-Direct

Labour

Materials

Re-curring

Non-Recurring

Plant

Sub-Contract

RISK AND OPPORTUNITY

RISK AND OPPORTUNITY

- **Risk:** The chance that your project will cost more or take longer than you estimated due to unknowns or uncertainties.
- **Opportunity:** The chance to reduce costs, increase margins, or add value during the project.

ESTIMATE = CONTROLLED UNCERTAINTY

Every estimate contains assumptions — things we believe to be true but can't fully guarantee.

Risk and opportunity management is how we stay realistic and competitive in the face of that uncertainty.

MAJOR SOURCES OF RISK

Unclear or incomplete scope

- Inadequate design or missing details in tender documents.
- Client expectations not fully defined.
- Risk: You underprice work that should have been excluded or clarified.

MAJOR SOURCES OF RISK

Material/subcontract price escalation

- Volatile markets (e.g., steel, concrete, fuel).
- Long project durations where future prices are uncertain.
- Risk: Supplier price increases after bid submission.

MAJOR SOURCES OF RISK

Productivity assumptions

- Overestimating how fast crews can work.
- Not accounting for site access issues, weather, or complexity.
- Risk: Labour and plant overruns.

MAJOR SOURCES OF RISK

Subcontractor risk

- Subcontractor under-pricing or going broke.
- Incomplete or misleading quotes.
- Risk: You wear the cost of their mistakes or failures.

MAJOR SOURCES OF RISK

Site conditions

- Unknown ground conditions (e.g., rock, groundwater).
- Environmental restrictions or contamination.
- Risk: Higher excavation, disposal, or compliance costs.

MAJOR SOURCES OF RISK

Contractual terms

- Unfavourable contract conditions (e.g., liquidated damages, performance guarantees).
- Delayed payments, retention clauses.
- Risk: Cash flow issues, penalties.

MAJOR SOURCES OF OPPORTUNITY

Value engineering

- Proposing alternative materials or methods.
- Example: Precast instead of in-situ concrete.
- Opportunity: Lower costs, higher margins.

MAJOR SOURCES OF OPPORTUNITY

Bulk procurement or Supplier agreements

- Discounts for larger orders or early commitments.
- Opportunity: Reduced material/unit rates.

MAJOR SOURCES OF OPPORTUNITY

High productivity/Efficient execution

- Optimised sequencing, experienced crews.
- Favourable weather, minimal delays.
- Opportunity: Lower labour/plant costs.

MAJOR SOURCES OF OPPORTUNITY

Client Relationships

- Repeat business may have lower risk or negotiation flexibility.
- Early access to information for better planning.

MAJOR SOURCES OF OPPORTUNITY

Early Completion Bonus

- Contracts with incentives for early delivery.
- Reduced indirect costs by finishing faster
- Opportunity: Additional revenue for efficient delivery.

HOW TO INCLUDE R&O

Percentage-based risk allowance

- Add a flat percentage (e.g., 5–10%) on top of the estimate.
- Simple but broad — suitable for early estimates.

Risk Build-Up (Bottom-Up Analysis)

- Identify individual risks, estimate impact $\times$ probability.
- Build a risk allowance based on specific issues.
- Allows more accurate, defensible pricing.

PROFIT AND OVERHEADS

PROFIT AND OVERHEADS

Up until now, estimating has been largely technical:

- Quantities measured
- Unit rates applied
- Risks analysed

But when it comes to Profit and Corporate Overheads, we move into business strategy.

PROFIT AND OVERHEADS

Where estimating moves from science to strategy

CORPORATE OVERHEADS

- These are the costs of running your business, not just a single project.
- They include:
 - Office rent, admin staff salaries, accounting, HR.
 - Company vehicles, IT systems, insurance.
 - Marketing, legal, and executive salaries.
- Overheads are typically recovered by allocating a percentage across all projects.
 - Example: If your annual overheads are \$2 million and you forecast \$20 million in revenue, you need 10% overhead recovery on every job to cover costs.

CORPORATE OVERHEADS

How to Apply Overheads in Estimates:

- Calculate the required overhead recovery rate.
- Apply this rate to direct + indirect costs to ensure the business is sustainable.

PROFIT

- Profit is the money you aim to make after all costs (direct, indirect, overhead) are covered.
- It is a pure business decision based on:
 - How much risk you're taking.
 - How much competition you face.
 - How much backlog you already have.
 - Your company's financial goals.

PROFIT - KEY QUESTIONS

How much do you want to win the work?

- Desperate for pipeline? Lower margin.
- Selective, premium contractor? Higher margin.

PROFIT - KEY QUESTIONS

How competitive is the market?

- High competition = pressure to lower margin.
- Niche or specialized work? You can justify more.

PROFIT - KEY QUESTIONS

What margins can you justify for the risk?

- High-risk jobs require higher margins.
- Low-risk, repeat-client jobs might justify less.

TYPICAL PROFIT MARGINS

- Tier 1 Contractors: 7% – 10%
- Mid-Tier Contractors: 10% – 15%
- Specialist/Niche Contractors: 10% – 20% (sometimes more)

PROFIT AND OVERHEADS

- Estimating is the scientific approach to calculating costs
- Profit and overheads are business decisions
- Great estimators need to understand both the scientific approach and the business decisions to win work

REVIEWING AND SUBMITTING

REVIEWING AND SUBMITTING

- Final step in the process - review and submit your estimate.
- Under-quoting jobs can send businesses broke
 - Never lose money
 - Better to have no work then to lose money
- Reviewing your estimate provides protection and commercial alignment

DOCUMENT AND QUALIFY

- An estimate is only valid if it reflects what you believe the job involves, but clients may have different expectations.
- That's why you must clearly document:
 - What's included in your price
 - What's excluded (scope you're not pricing).
 - Any assumptions (e.g., access to site, working hours, material availability).

LETTER OF OFFER

- Every estimate should be accompanied by a Letter of Offer or Bid Clarifications, including:
 - Scope summary: What your price covers.
 - Key exclusions: Items not priced.
 - Assumptions: Any conditions you've based your price on.
 - Terms & Conditions: Payment terms, validity of offer, escalation clauses.

TERMS AND CONDITIONS

- Your price is based on your understanding of the contract.
- If the contract changes after submission (e.g., more risk, tighter deadlines), the price must change.
- Always align your price to your commercial terms — or you risk pricing for one thing, and being forced to deliver something else.

DETAILED REVIEW

- Not all parts of your estimate are equally important — apply the 80/20 Rule:
 - 80% of your cost is likely tied to 20% of the items.
 - Focus your detailed review on the big-ticket costs.

DETAILED REVIEW

Key Areas to Review Critically:

- Major cost items: Concrete, steel, plant, labour hours.
- Subcontractor quotes: Are they aligned with your scope?
- Productivity assumptions: Are they realistic?
- Risk allowances: Are they sufficient, or padded?
- Indirect costs: Have you covered the full duration?

STRATEGIES

High-Level Sanity Check:

- Compare total cost/m² or \$/W to previous similar jobs.
- Does the total price feel right? If not, dig deeper.

STRATEGIES

Work Package Review:

- Are all scope items included?
- Any duplication or omissions?

STRATEGIES

Mathematical Check:

- Verify formulas, summaries, and unit conversions.
- Small errors here can cause big mistakes.

STRATEGIES

Program Alignment:

- Are costs and indirects aligned with the construction program?

STRATEGIES

Always have someone else review your estimate before submission.

- A fresh set of eyes can catch:
 - Calculation mistakes.
 - Scope gaps.
 - Misinterpretations of contract terms.

SUBMISSION

- Ensure the submission includes:
 - Final estimate summary.
 - Letter of Offer with all assumptions, exclusions, and terms.
 - Supporting documents (BOQ, schedules, program if required).
 - Contact details for clarification.

CONCLUSION